Day 14: Multicollinearity

Today, we're going to explore what happens when you are using multiple predictors that are correlated with one another.

Example 1: Uncorrelated Predictors

As a first example, let's consider a multiple linear regression model where the two predictor variables are uncorrelated.

Task 1. Inspect the below correlation matrix and verify that the correlation between the two predictor variables (x1 and x2) is 0.

```
          x1        x2         y
x1 1.0000000 0.0000000 0.7419309
x2 0.0000000 1.0000000 0.6384057
y  0.7419309 0.6384057 1.0000000
```

Task 2. The below code chunk fits a simple linear regression model with y as the response variable and x1 as the predictor variables. The coefficients table was extracted using the `broom::tidy()`. (Feel free to use `broom::tidy()` in your work!)

```
# A tibble: 2 x 5
  term        estimate std.error statistic p.value
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)    23.5      10.1       2.32  0.0591
2 x1              5.37      1.98      2.71  0.0351
```

Similarly, here is the output for the simple linear regression model with y as the response variable and x2 as the predictor variable.

```
# A tibble: 2 x 5
  term        estimate std.error statistic p.value
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)    27.3      11.6       2.35  0.0572
2 x2              9.25      4.55      2.03  0.0885
```

And finally, here is the output for the multiple linear regression model with y as the response variable and both x1 and x2 as the predictor variables.

```
# A tibble: 3 x 5
  term        estimate std.error statistic  p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)    0.375     4.74     0.0791 0.940
2 x1             5.37      0.664    8.10   0.000466
3 x2             9.25      1.33     6.97   0.000937
```

Compare the model summaries from the three models. What do you observe about the coefficients when both predictors are included in the model? What do you observe about the standard errors? The results of the hypothesis tests?

Example 2: Perfectly Correlated Predictors

In many examples the predictor variables will be correlated. As an extreme example, let's consider a situation where the predictors are perfectly correlated. The data set for this small example is printed below.

```
  x1 x2   y
1  2  6  23
2  8  9  83
3  6  8  63
4 10 10 103
```

Task 3. Inspect the below correlation matrix and verify that the correlation between the two predictor variables (x1 and x2) is 1.

```
   x1 x2 y
x1  1  1 1
x2  1  1 1
y   1  1 1
```

Task 4. From the previous task, you should have seen that the correlation between x1 and x2 is 1, as is the correlation between both predictors and y. This sounds like a great situation, right? We'll explore two models to unpack this.

a. Consider the fitted regression equation: $\widehat{y} = -87 + x_1 + 18x_2$. Verify that the predicted values are exactly equal to the observed values.

x1	x2	y	yhat
2	6	23	
8	9	83	
6	8	63	
10	10	103	

b. Consider the fitted regression equation: $\widehat{y} = -7 + 9x_1 + 2x_2$. Verify that the predicted values are exactly equal to the observed values.

x1	x2	y	yhat
2	6	23	
8	9	83	
6	8	63	
10	10	103	

c. Discuss the following questions with your group:

- Would you be willing to use either model to make predictions for new observations? Why or why not?

- Would you be willing to interpret the coefficients in either model? Why or why not?

- Do you think the standard error of the slopes is reasonably small or large? Why?

Example 3: A more realistic example

Let's consider a more realistic example where the predictor variables are correlated, but not perfectly correlated. The data set `bodyfat` contains measurements on 20 healthy adults between 25 and 34 years of age. We're interested in predicting body fat percentage using three body measurements: triceps skinfold thickness, thigh circumference, and midarm circumference.

Task 5. Below is the correlation matrix for the data set. What do you observe about the correlations between the predictor variables?

	triceps_skinfold	thigh_circumference	midarm_circumfer-ence	body_fat
triceps_skinfold	1.000	0.924	0.458	0.843
thigh_circumference	0.924	1.000	0.085	0.878
midarm_circumfer-ence	0.458	0.085	1.000	0.142
body_fat	0.843	0.878	0.142	1.000

Task 6. There are numerous models that we could consider. Below are summary tables from several models. Compare the model summaries from the fitted models. What do you observe about the coefficients when different predictors are included in the model? What do you observe about the standard errors? The results of the hypothesis tests?

Only triceps skinfold as a predictor:

```
# A tibble: 2 x 5
  term             estimate std.error statistic    p.value
  <chr>               <dbl>     <dbl>     <dbl>      <dbl>
1 (Intercept)        -1.50      3.32     -0.451 0.658
2 triceps_skinfold    0.857     0.129     6.66  0.00000302
```

Only thigh circumference as a predictor:

```
# A tibble: 2 x 5
  term                estimate std.error statistic     p.value
  <chr>                  <dbl>     <dbl>     <dbl>       <dbl>
```

```
1 (Intercept)          -23.6       5.66      -4.18 0.000566
2 thigh_circumference    0.857     0.110      7.79 0.000000360
```

Only midarm circumference as a predictor:

```
# A tibble: 2 x 5
  term                estimate std.error statistic p.value
  <chr>                  <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)           14.7       9.10      1.61    0.124
2 midarm_circumference   0.199     0.327     0.611   0.549
```

Triceps skinfold and thigh circumference as predictors:

```
# A tibble: 3 x 5
  term               estimate std.error statistic p.value
  <chr>                 <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)          -19.2       8.36     -2.29   0.0348
2 triceps_skinfold       0.222     0.303     0.733  0.474
3 thigh_circumference    0.659     0.291     2.26   0.0369
```

Triceps skinfold and midarm circumference as predictors:

```
# A tibble: 3 x 5
  term                estimate std.error statistic     p.value
  <chr>                  <dbl>     <dbl>     <dbl>       <dbl>
1 (Intercept)          -26.0       7.00     -3.72  0.00172
2 midarm_circumference   0.0960    0.161     0.595 0.560
3 thigh_circumference    0.851     0.112     7.57  0.000000772
```

Thigh circumference and midarm circumference as predictors:

```
# A tibble: 4 x 5
  term                estimate std.error statistic p.value
  <chr>                  <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)           117.      99.8       1.17   0.258
2 triceps_skinfold        4.33     3.02      1.44   0.170
3 thigh_circumference    -2.86     2.58     -1.11   0.285
4 midarm_circumference   -2.19     1.60     -1.37   0.190
```

Let me know when you have reached this spot. We'll regroup as a class to discuss your observations.

Multicollinearity

Multicollinearity is the situation where two or more predictors are "nearly" perfectly correlated with each other.

Diagnosing multicollinearity

- Examine a scatterplot matrix
- Examine pairwise correlations between predictors
- Look for $\widehat{\beta}_i$'s with unusual _____ (opposite of what you'd expect)
- Notice sensitivity to changes in the model / order of predictors
- Calculate the **Variance Inflation Factor (VIF)**

> Variance Inflation Factor
>
> $$VIF_i = \frac{1}{1 - R_i^2}$$
>
> where R_i^2 = R^2 from a model predicting x_i using *all other* predictor variables (not y!).
> **Guidelines:**
>
> | VIF_i | R_i^2 | Interpretation |
> |---|---|---|
> | > 5 | $> .8$ | suspicions begin |
> | > 10 | $> .9$ | indicates a problem |
> | > 100 | $> .99$ | indicates a big problem |

VIFs for first example are modest – nothing to worry about.

```
ex1_mod <- lm(y ~ x1 + x2, data = ex1)
car::vif(ex1_mod)
```

```
x1 x2
 1  1
```

VIFs for the second example:

```
ex2_mod <- lm(y ~ x1 + x2, data = ex2)
car::vif(ex2_mod)
```

```
Error in vif.default(ex2_mod): there are aliased coefficients in the model
```

`car::vif()` throws an error! Why?

Since x_1 and x_2 are perfectly correlated here, R_i^2 =***, so the denominator*** $1 - R_i^2$ = and VIF = _____.

Example 3: real data (`bodyfat`)

```
bodyfat_mod <- lm(body_fat ~ triceps_skinfold + thigh_circumference + midarm_circumference,
                  data = bodyfat)
car::vif(bodyfat_mod)
```

```
    triceps_skinfold  thigh_circumference midarm_circumference
            708.8429              564.3434              104.6060
```

Thigh and triceps skinfold circumference are both driven largely by overall body fat, so they move together.

What about predictions?

Now that we've discussed the issues revolving around interpreting coefficients and drawing inference on them in the presence of multicollinearity, let's consider how multicollinearity impacts predictions.

Recall that the mean squared error (MSE) of a model measures the variability in the error terms (residuals). In other words, it is measuring the variability unexplained by the model.

Task 7. Below are the MSE values for three of the models we considered in Task 6. What do you observe about the MSE values when different predictors are included in the model?

Predictors	MSE
triceps_skinfold	7.95
triceps_skinfold, thigh_circumference	6.47
triceps_skinfold, thigh_circumference, midarm_circumference	6.15

Task 8. Based on your observations, do you think we can use any of the models to make predictions for new observations? Why or why not?

Let me know when you have reached this spot. We'll regroup as a class to discuss your observations.

Remedial measures

- Use the model only for *prediction*, if you believe future observations will preserve the same relationships among predictors
- For polynomial terms, use `poly(x, degree)` in R, or center the variable
- Drop some of the highly correlated variables (use caution, this doesn't always work and loses information)
- Add cases that break the correlation between predictors – reasonable in designed experiments
- Create composite variables (sums, averages, principal components) – can fix collinearity but be harder to interpret
- Use ridge regression or LASSO – covered in Stat 270 (Statistical Learning)